

Sourabh Ranjan Kar Full Stack Developer

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Summary

Full-Stack Developer with 3+ years of experience in fintech and logistics, building scalable and secure applications that improve performance and user experience. Skilled in Java/Spring Boot for backend services and React/TypeScript for developing responsive interfaces. Experienced in REST API design, CI/CD automation, and deploying microservices on AWS ECS (Fargate) using CloudFormation and Lambda integrations. Proficient in PostgreSQL and Docker, with hands-on experience in Python, applied ML, and Generative-AI integrations (OpenAI API, LangChain).

Technical Skills

- Frontend:** JavaScript (ES6+), React.js, Angular, Redux, TypeScript, Responsive UI/UX
- Backend & Cloud:** Java (Spring Boot, REST APIs, Microservices, JPA/Hibernate), AWS (ECS, Lambda, API Gateway, RDS, CloudFormation), Docker, Kubernetes
- Databases:** PostgreSQL, MySQL, Azure SQL, MongoDB
- Testing & Code Quality** JUnit, Mockito, Postman, Jest, SonarQube
- Security:** Spring Security, OAuth2, JWT
- AI & GenAI:** Python (Pandas, NumPy, scikit-learn), OpenAI API, LangChain, Prompt Engineering, LLM Integration (GPT, Claude), Predictive Modeling, Regression, Clustering, Data-Driven APIs
- Practices:** Agile/Scrum, Event-Driven Architecture (Kafka/SQS), CI/CD Automation

Professional Experience

- Full Stack Developer, Fifth Third Bancorp**

10/2024 – Present | Remote, USA

 - Engineered the KYC verification and onboarding dashboard (React/TypeScript + Spring Boot microservices), integrating LexisNexis APIs for AML checks and enhancing completion accuracy while reducing setup time by 20%.
 - Designed and enhanced key REST APIs (Spring Boot/JPA behind AWS API Gateway) by adding pagination, idempotency, and caching, improving reliability and performance for high-volume workloads.
 - Optimized AWS RDS with partitioning and indexing, boosting query performance by 40% and accelerating dashboard responsiveness for compliance reporting.
 - Introduced event-driven workflows using AWS SQS to process high-volume onboarding tasks asynchronously, decoupling services and reducing response latency by 25% under peak loads.
 - Automated infrastructure provisioning with AWS CloudFormation (API Gateway, Lambda, RDS) and set up S3 presigned uploads for KYC documents with lifecycle policies and IAM-scoped access, reducing environment setup time and strengthening file security.
 - Deployed Dockerized microservices on AWS ECS (Fargate) with blue/green rollouts via CodePipeline and added CloudWatch/X-Ray monitoring, moving releases from weekly to daily and reducing MTTR.
- Full Stack Developer, Federal Express Corporation AMEA (FedEx)**

01/2021 – 07/2023 | India

 - Delivered shipment-tracking modules for AMEA logistics using Angular and Spring Boot, enabling real-time tracking across 20+ countries and improving shipment visibility by 30% across international operations.
 - Created a multilingual Angular front end using Angular Material, ngx-translate, and reactive forms, where A/B tests improved engagement and usability, enhancing accessibility and adoption across APAC and EMEA regions.
 - Implemented real-time shipment updates with WebSockets, RxJS, and NgRx, integrating with backend microservices powered by Kafka to ensure reliable event streaming and synchronized state for time-critical deliveries.
 - Refined Azure SQL (stored procedures, indexing, partitioning) in collaboration with the database team to handle millions of shipment records, cutting query response times by 40%.
 - Strengthened backend security by integrating Spring Security, OAuth2, JWT, RBAC into Spring Boot services; reduced vulnerabilities by 25% following pen-test remediation.
 - Built API and service-level test suites (JUnit, Mockito, Postman), expanding coverage and reducing defect leakage to enhance production stability.
 - Collaborated with DevOps engineers to enhance CI/CD in Azure DevOps with SonarQube checks, automated tests and gated approvals, reducing deployment cycles from 2 days to 6 hours and improved release success rate by 20%.

Projects

- Generative AI Knowledge Assistant | Personal Project**

Python, LangChain, OpenAI API, Streamlit, AWS Lambda, PostgreSQL

 - Developed a prototype AI assistant that summarizes documents and answers questions using GPT models through LangChain.
 - Integrated embeddings and OpenAI GPT API to enable semantic search and automated insight generation.
 - Deployed serverlessly on AWS Lambda, achieving sub-second query responses and improving internal knowledge accessibility.
- Predictive Modeling on Alzheimer's Disease and Healthy Aging | MS Capstone, University of North Texas**

Python, Pandas, NumPy, Scikit-learn, Random Forest, Linear Regression, K-Means, Tableau

 - Built regression and classification models to analyze CDC aging data and identify key factors influencing cognitive health.
 - Applied Random Forest and K-Means clustering to uncover state-level health behavior patterns and segment populations by risk.
 - Visualized findings in Tableau to highlight preventive healthcare trends and mental well-being correlations across U.S. states.

Education

- University of North Texas, Denton, TX**

Master of Science in Advanced Data Analytics

08/2023 – 05/2025
- Loyola Academy Degree and PG College, India**

Bachelor of Science Computer Science and Engineering

06/2017 – 11/2020

Certificates

- AWS Certified Developer - Associate**

GenAI and LLMs on AWS